Introduction to Bioinformatics: A Computer Science Perspective

Main References:

- **Visual Guide:** Concepts of Biology (OpenStax)
- **Course Gateway:** CENG465 - Introduction to Bioinformatics

Outline: What is Bioinformatics? $\longrightarrow$ The Biological Data Stack $\longrightarrow$ Tasks and Algorithms

What is Bioinformatics?

(Molecular) **Bio** - **informatics**

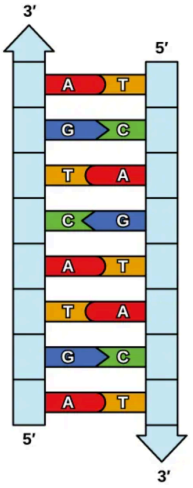
One idea for a definition?

> **Bioinformatics** is **conceptualizing biology in terms of molecules** (in the sense of physical-chemistry)
>
> and then applying **"informatics" techniques** (derived from disciplines such as applied math, CS, and statistics)
>
> to **understand** and **organize** the **information** associated with these molecules, **on a large-scale**.
>
> **Bioinformatics** is a practical discipline with many **applications**.

To a computer scientist, the cell is not just a biological unit; it is a complex, hierarchical **Computing System**. To analyze this system, we must first define its **Data Stack**—the layers of information that flow from persistent storage to functional execution.

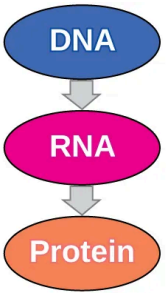
The Biological Data Stack

To analyze this system, we must map biological molecules to familiar computational roles. Just as a computer transitions from disk storage to runtime memory and finally to binary execution, the cell manages its operations through a hierarchical data stack.



DNA | Persistent Storage (Source Code)

- **Format:** String over $\{A, T, C, G\}$.
- **Properties:** High redundancy; includes non-coding "comments"; managed via evolutionary version control.
- **Example:** `5'-TCAGTACT-3'` (Directional).



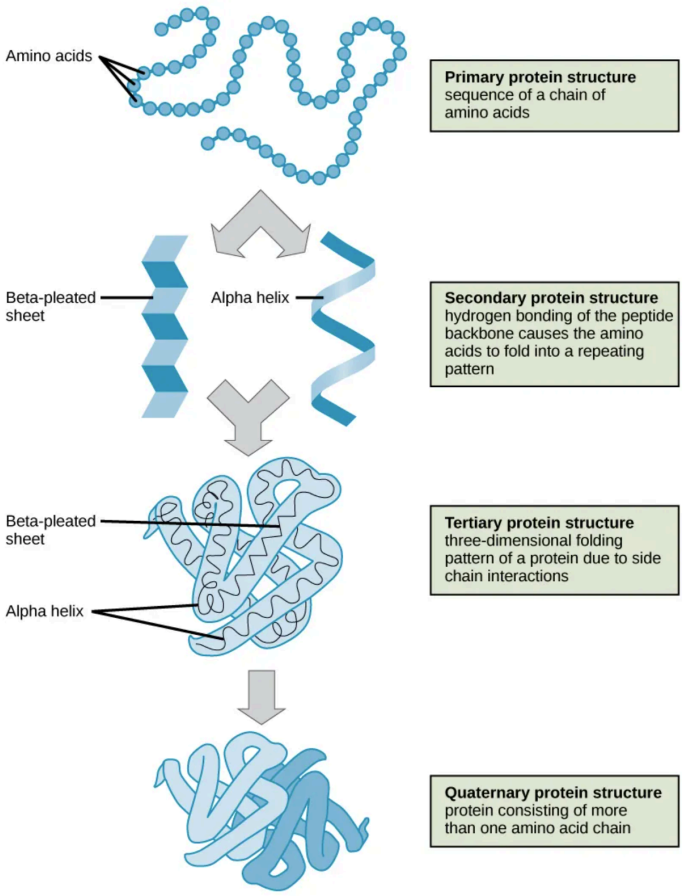
RNA | Runtime Snapshot (Middleware)

- **Format:** String over $\{A, U, C, G\}$.
- **CS View:** Volatile "copy" of a code snippet (Transcript).
- **Data Representation:** A state vector in $\mathbb{R}^{20,000}$ (Expression levels across the genome).

Protein | Functional Binary (Execution)

Proteins fold from 1D strings into 3D machines through four data layers:

- **Primary (1°): 1D Array**. A linear sequence of amino acids with fixed directionality (N to C-terminus).
- **Secondary (2°): Local Subroutines**. Recurring patterns like α-helices and β-sheets.
    - *Search Space:* Restricted by (ϕ, ψ) angles (Ramachandran Plot).
- **Tertiary (3°): 3D Object**. The full geometric topology.
    - *Data Structure:* 3D Point Cloud or **Contact Map** (Adjacency Matrix).
- **Quaternary (4°): Multi-core Assembly**. Interaction of multiple protein units (e.g., Dimers/Tetramers).



Biological Networks | System Call Graph

- **Format:** Heterogeneous Graph $G = (V, E)$.
- **Nodes (V):** Proteins, Genes, Metabolites.
- **Edges (E):** Interactions (Binding, Control, Metabolic flux).

Tasks and Algorithms

To decode the biological stack, we transition from **Pattern Matching** (Classics) to **System Simulation** (Frontier). This paradigm shift marks the move from heuristic-based lookup to representation-based inference.

The following examples illustrate these foundational tasks and their algorithmic evolution:

Data Layer	Primary Task	Classic Model (Heuristics)	AI Frontier Model
DNA	**Homology Detection**	**SW-DP / BWT** Needleman, 1970	**HyenaDNA** Poli et al., 2023
RNA	**Cell State Mapping**	**Leiden Clustering** Traag, 2019	**scGPT** Cui et al., 2024
Protein	**3D Folding**	**Profile HMM** Eddy, 2011	**AlphaFold 3** Abramson et al., 2024
System	**Workflow Discovery**	**GSEA** Subramanian, 2005	**GNN / DeepPurpose** Huang et al., 2022

As a conclusion, Bioinformatics represents the **Reverse Engineering** of life's source code.

> **"What I cannot create, I do not understand." —— Richard Feynman**